

Perception Contracts for Reasoning about Reliability of Visual Autonomy: Incremental Stability Certificates

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We present a unified verification framework for certifying *incrementally stable perception contracts (ISPC)*. A perception contract provides an environment-aware, set-valued abstraction of a learned vision pipeline; composing it with the plant dynamics yields a set-valued system amenable to formal analysis. We verify incremental stability by finding a quadratic contraction metric and rate satisfying a uniform matrix inequality over admissible Jacobians induced by the perception contract. This search is formulated as bisection over the contraction rate with semidefinite feasibility checks at each step, producing certified bounds on convergence rates and the diameter of the ultimately invariant tube. As a special case, when the system admits an equilibrium, incremental stability implies *asymptotic stability*. For this case, we introduce *Lyapunov perception contracts (LPCs)*, which combine perception contract conformance with Lyapunov-based correctness guarantees and enable operating design domain (ODD) identification in finitely parameterized environments. We also present a method for computing perception contracts through abstract rendering, which propagates pose and scene uncertainty through photorealistic rendering pipelines. We demonstrate the framework on two vision-based control tasks: formation flying with 6D rigid-body dynamics using GateNet and UNet+PnP perception pipelines in photorealistic Unreal Engine simulations, and automated landing (AUTOLAND) using rendered and real-world imagery. Experimental results show that quadratic perception contracts tighten certified tubes by 30–35% over linear contracts, and that GateNet-based estimators achieve faster verified convergence than UNet+PnP. These results demonstrate that perception contracts paired with incremental stability analysis provide rigorous system-level guarantees for vision-based control under high-dimensional visual uncertainty.

CCS Concepts: • **Computing methodologies** → **Artificial intelligence**; • **Computer systems organization** → **Embedded and cyber-physical systems**; • **Networks** → *Network safety and reliability*.

Additional Key Words and Phrases: Incremental Stability, Perception Contracts, Lyapunov Theorem, Set-valued Systems, Visual Control

1 Introduction

Visual autonomous systems rely on learned perception pipelines to estimate state from images, which then drive control decisions. Such vision-in-the-loop architectures now appear in safety-critical domains including automated landing [24, 40, 42], formation flight [7, 15], and driver assistance [16, 33]. Assuring reliability in these systems is challenging: even in scenarios with identical geometry, small environmental variations (for example, in lighting, weather, background scene) can introduce large changes in pixel colors, that can degrade the performance of downstream state estimators and controllers.

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Consider a closed-loop system in which a learned perception pipeline—also called an *observer*—drives the controller. To enable modular reasoning about reliability of such visual autonomous systems, we present the *perception contract* framework [4, 16, 24]. A perception contract over-approximates the perception pipeline’s outputs as a set-valued map over a specified environment, guaranteeing a system-level correctness requirement such as safety or stability. Composing this set-valued map with the plant dynamics yields an abstract closed-loop system whose trajectories conservatively bound those of the original vision-based system. Verifying properties of this abstraction therefore yields sound guarantees—without requiring differentiable models of the learned perception pipeline.

A key application of perception contracts is identifying *operating design domains* (ODDs), the environmental conditions under which a vision-based system can be certified safe. Environmental factors such as lighting direction, weather conditions, and scene complexity directly affect perception accuracy, which in turn impacts closed-loop stability. Our framework enables systematic identification of a maximal subset of environmental parameters under which formal stability guarantees hold, providing actionable boundaries for safe deployment of vision-based autonomous systems.

A key system-level property we verify is *incremental stability* [2], which requires that the separation between any two trajectories of the closed-loop system contracts over time or remains uniformly bounded despite disturbances. Incremental stability is a general robustness property applicable to systems with or without equilibria, and captures essential performance requirements for vision-based tracking, formation maintenance, and trajectory following. Notably, incremental stability is *more general* than asymptotic stability: it applies to time-varying reference trajectories and does not require convergence to a single equilibrium point.

We show how perception contracts for incremental stability can be constructed using *contraction analysis* [27]. Given a contract, we compute a contraction metric and rate via a uniform matrix inequality over the induced Jacobian set. When the contraction rate is negative, the system is incrementally stable with trajectories converging to an ultimately invariant tube. We formulate the search for contraction rates as semidefinite programs solved via bisection, yielding tight, computable bounds on convergence rates and ultimate tube diameters.

When the system has an equilibrium, incremental stability specializes to *asymptotic stability*—convergence to a target set. For this case, our prior work [24] introduced *Lyapunov perception contracts* (LPCs), which combine conformance guarantees (the contract bounds the observer) with correctness guarantees (a Lyapunov function certifies stability). LPCs enable ODD identification through an iterative refinement procedure: we construct perception contracts over progressively smaller environmental regions, verify Lyapunov decrease conditions, and shrink the environment when verification fails. The resulting ODD represents the largest certified safe operating region.

In Section 4.7, we discuss methods for constructing perception contracts through multiple approaches. Statistical methods sample the perception pipeline under varied environments and fit set-valued bounds via regression [16, 24]. Alternatively, *abstract rendering* [17] propagates pose and scene uncertainty through the rendering process (for example for Gaussian Splatting and NeRFs) to compute *abstract images* which can then be passed through a set-valued observer. While abstract rendering provides tight contracts when scene reconstructions are available, our framework addresses the general case of black-box ML-based observers, enabling verification without explicit scene models. Both methods integrate seamlessly with our incremental stability analysis.

We illustrate the perception contract framework on two realistic visual autonomous systems. First, we evaluate formation flying with six-degree-of-freedom rigid-body aircraft dynamics using GateNet [34] and UNet+PnP [23, 37] perception pipelines in photorealistic Unreal Engine environments under diverse lighting, weather, and background conditions. Experiments illustrate that quadratic perception contracts shrink the ultimately invariant regions by approximately 30–35% over linear contracts while tightening the contraction rate. We observe that GateNet contracts

are consistently tighter than UNet+PnP contracts for identical dynamics and environments. Additionally, enlarging the environmental domain nearly doubles the invariant diameter and pushes the convergence rate toward instability, whereas relaxing the confidence level from 99% to 95% appreciably tightens the tube diameter while lowering statistical coverage. Second, we demonstrate automated landing using images from both rendered simulations and Google Earth under varying lighting and scene conditions, certifying operating design domains through Lyapunov perception contracts. These results provide direct reliability guarantees: a tighter ultimately invariant set certifies smaller worst-case deviation from the reference trajectory, and a more negative contraction rate indicates faster recovery from disturbances. The certified ODD precisely identifies which environmental conditions (lighting, weather, background) admit formal stability guarantees, providing operators with provable bounds on closed-loop behavior under visual uncertainty.

The main contributions of this paper are as follows.

- (1) We develop *incrementally stable perception contracts (ISPC)*, a unified framework for certifying incremental stability of vision-based closed-loop systems under perception uncertainty. Perception contracts allow the concrete system to be over-approximated by a tractable abstract system. We prove that incremental stability of the abstract system guarantees that of the concrete system, with computable bounds on trajectory deviation and ultimate invariant tube diameter (Theorem 1, Section 4).
- (2) We provide efficient algorithms for constructing perception contracts from data and computing certified contraction rates. Contract learning is posed as a regression problem over sampled observations, while contraction rate computation is reduced to bisection over semidefinite feasibility problems, yielding tight certified bounds on convergence rate and worst-case deviation (Algorithms 1 and 2).
- (3) We validate the framework on formation flying with GateNet and UNet+PnP perception pipelines in Unreal Engine environments, and on automated landing using Gazebo and Google Earth data, demonstrating practical incremental stability certificates under realistic visual uncertainty (Section 5).

Organization

The remainder of the paper is organized as follows. Section 2 surveys related work on verification techniques, perception contracts, and incremental stability. Section 3 introduces the problem formulation, covering dynamical systems and incremental stability with contraction metrics, followed by asymptotic stability with Lyapunov functions as a special case. Section 4 presents our technical contributions: the construction of perception contracts, the incremental stability framework with contraction rate computation, and Lyapunov perception contracts for asymptotic stability and ODD identification. Section 5 details experimental evaluation on automated landing and formation flying. Section 6 concludes and discusses future directions.

2 Related Work

Incremental Stability and Contraction Analysis. Classical incremental stability theory provides Lyapunov-based criteria ensuring that distances between trajectories decay over time [2]. Contraction theory offers an equivalent Jacobian-based characterization, requiring uniformly negative matrix measures of the generalized Jacobian [27]. A quadratic Lyapunov function of differential states, defined by a Riemannian contraction metric, gives necessary and sufficient conditions for incremental exponential convergence [44]. Recent tutorials [18, 44] show how contraction metrics can be constructed via convex optimization, yielding explicit exponential bounds on trajectory deviations. Our work extends this framework to systems with environment-dependent observers abstracted by perception contracts.

Verification of Vision-Based Autonomous Systems. Formal analysis of vision-based autonomy has gained significant attention; see [31] for a comprehensive survey. Applications span driving [4, 8, 16], automated landing [24, 39, 43], and multi-agent coordination [15]. Shoukry and collaborators developed verified landing methods using geometric camera models: NNlander-VeriF [39] converts camera geometry into neural networks for closed-loop verification, subsequent work [6] exploits mixed monotonicity for certifiable state-estimation error bounds, and [22] extends this to certified 3D pose estimation via reachability analysis. To date, these efforts have focused on safety and reachability rather than stability.

Conformal Prediction for Perception Uncertainty. Conformal prediction [41] provides statistical bounds on perception error and has been applied to runtime monitoring [26], navigation [29], and planning [25]. Recent extensions integrate conformal prediction with control barrier functions for safety under imperfect perception [45], and state-dependent variants [14] exploit state-error correlations for tighter reachability bounds. However, conformal methods do not directly yield set-valued abstractions for system-level stability; perception contracts fill this gap through compositional verification.

Perception Contracts. In our prior work, we introduced perception contracts—set-valued abstractions of ML-based observers that enable compositional reasoning about downstream control and plant dynamics [4, 16], with extensions to multi-agent coordination [15]. We developed learning-based construction methods [4, 16, 24] that train neural network approximations of vision pipelines and compute over-approximations via interval bound propagation or abstract interpretation, handling black-box observers without explicit scene models. More recently, we introduced abstract rendering [17], which propagates camera pose uncertainty through photorealistic pipelines (Gaussian Splatting, NeRF), decomposing rendering into linearly over-approximable operations to maintain tight bounds. The present paper builds on both construction approaches.

This work advances the state of the art in three ways: (i) *incrementally stable perception contracts (ISPC)*, a unified framework that proves incremental stability of the abstract system guarantees that of the concrete system, with computable bounds on trajectory deviation and invariant tube diameter; (ii) efficient algorithms for perception contract learning from data and contraction rate computation via bisection over semidefinite feasibility problems, yielding tight certified bounds on convergence and worst-case deviation; and (iii) experimental validation on formation flying with GateNet and UNet+PnP pipelines and automated landing using Gazebo and Google Earth data, demonstrating practical stability certificates under realistic visual uncertainty.

3 Overview of Incremental Stability with Perception Contracts

3.1 Preliminaries

Notations. For a real symmetric matrix $A \in \mathbb{R}^{n \times n}$, $A \succ 0$ ($A \prec 0$) indicates that A is positive (negative) definite. The set of positive definite $n \times n$ matrices is denoted by $\mathbb{S}_{>0}^n$. For $P \in \mathbb{S}_{>0}^n$, the P -norm of a vector $x \in \mathbb{R}^n$ is defined as $\|x\|_P := \sqrt{x^\top P x}$. The maximum and minimum eigenvalues of $P \in \mathbb{S}_{>0}^n$ are denoted by $\lambda_{\max}(P)$ and $\lambda_{\min}(P)$. The set $E_{P,c}(x^*) = \{x \in \mathbb{R}^n \mid \|x - x^*\|_P \leq c\}$ denotes the ellipsoid centered at x^* with shape P and size c ; and $B_c(x^*) = E_{I,c}(x^*)$ represents the Euclidean ball centered at x^* of radius c . For a bounded set $X \subset \mathbb{R}^n$, the *diameter* of X is defined as the maximum distance between any two points in the set: $d_P(X) = \sup_{x,y \in X} \|x - y\|_P$. The *distance from a point x to a set X^** in the P -norm is defined as $d_{P,X^*}(x) = \inf_{y \in X^*} \|x - y\|_P$. For a smooth function $f : X \rightarrow Y$, its Jacobian is denoted by $J_f := \frac{\partial f}{\partial x} \in \mathbb{R}^{m \times n}$.

We will use set-valued functions for modeling the evolution of systems with uncertain perception. For a set-valued function $F : X \rightarrow 2^Y$, a *smooth cover*, if one exists, is a family of smooth functions $\mathcal{F}$ such that $F(x) = \{f(x) | f \in \mathcal{F}\}$ for all $x \in X$. A member $f \in \mathcal{F}$ is called a *smooth selection* of F .

Given a compact set $X^* \subseteq X$ and a smooth cover $\mathcal{F}$ of $F : X \rightarrow 2^Y$, we define the *Jacobian set of F over X^** as:

$$\mathcal{J}_{X^*} = \{J_f(x) \in \mathbb{R}^{m \times n} \mid x \in X^*, f \in \mathcal{F}\}.$$

Finally, the *point-wise P -diameter* of F over X^* is defined as:

$$C_{P,X^*} = \sup_{f_1, f_2 \in \mathcal{F}, x \in X^*} \|f_1(x) - f_2(x)\|_P.$$

We conclude this preliminary subsection with a several basic facts about Jacobian sets and point-wise diameters. The first proposition formalizes a standard property: the Jacobian set of F over a convex set X^* is closed under integration.

Proposition 1. *Let $X^* \subseteq X$ be compact convex, and $\mathcal{F}$ be a smooth cover of $F : X \rightarrow 2^Y$. If $\mathcal{J}_{X^*}$ is convex, then for any $f \in \mathcal{F}$ and any $x_1, x_2 \in X^*$,*

$$\int_0^1 J_f(x_2 + s(x_1 - x_2)) ds \in \mathcal{J}_{X^*}.$$

PROOF. Because X^* is convex, the point $x_2 + s(x_1 - x_2)$ lies in X^* for all $s \in [0, 1]$. Therefore, $J_f(x_2 + s(x_1 - x_2)) \in \mathcal{J}_{X^*}$. By convexity of $\mathcal{J}_{X^*}$, any convex combination of its elements also lies in $\mathcal{J}_{X^*}$, and the integral $\int_0^1 J_f(x_2 + s(x_1 - x_2)) ds$ is precisely such a convex combination. Hence it lies in $\mathcal{J}_{X^*}$. $\square$

The next Proposition establishes the existence and finiteness of point-wise P -diameters and it follows immediately from continuity of norms and compactness of the domain.

Proposition 2. *Let $X^* \subseteq X$ be compact, and $\mathcal{F}$ be a smooth cover of set-valued map $F : X \rightarrow 2^Y$. For any matrix $P \in \mathbb{S}_{>0}^n$, C_{P,X^*} exists and is finite.*

PROOF. For each $f_1, f_2 \in \mathcal{F}$, the map $x \mapsto \|f_1(x) - f_2(x)\|_P$ is continuous on the compact set X^* , hence attains a finite maximum C_{f_1, f_2} . Defining $C_{P,X^*} := \sup_{f_1, f_2 \in \mathcal{F}} C_{f_1, f_2}$ gives a uniform bound satisfying $\|f_1(x) - f_2(x)\|_P \leq C_{P,X^*}$, $\forall f_1, f_2 \in \mathcal{F}, x \in X^*$. $\square$

The final Proposition of this section provides an upper bound on $x^T P y$ in terms of the quadratic terms $x^T P x$ and $y^T P y$, which will be useful later for incremental stability analysis.

Proposition 3. *For any $P \in \mathbb{S}_{>0}^n$, any $x, y \in \mathbb{R}^n$, and $\epsilon > 0$:*

$$2x^T P y \leq \epsilon x^T P x + \frac{1}{\epsilon} y^T P y.$$

PROOF. By the Cauchy-Schwarz inequality, $x^T P y \leq \|x\|_P \|y\|_P$, and applying the scalar Young's inequality $2ab \leq \epsilon a^2 + \frac{1}{\epsilon} b^2$ with $a = \|x\|_P$ and $b = \|y\|_P$ gives the result. $\square$

3.2 Incremental Stability in Set-Valued Dynamics

We will use continuous-time dynamical systems with set-valued dynamics to model uncertainty. In this section, we introduce the system model and provide the theorem for obtaining rigorous bounds on differences between their executions.

An n -dimensional continuous-time system over a state space $X \subseteq \mathbb{R}^n$ with set-valued dynamics is defined by

$$\dot{x} \in F(x), \quad (1)$$

where $F : X \rightarrow 2^X$ admits a smooth cover $\mathcal{F}$ and $X_0 \subset X$ is the set of initial states.

An *execution* of the system is an absolutely continuous function $x : [0, \infty) \rightarrow X$ such that $x(0) \in X_0$ and $\dot{x}(t) \in F(x(t))$ for all $t \geq 0$ at which $\dot{x}(t)$ exists. As the system can have multiple initial states and the dynamics are set-valued, it is nondeterministic, with possibly multiple executions.

The *reachable set* over time interval $[t_1, t_2]$ is the set of all states in any execution between and including t_1 and t_2 :

$$\text{Reach}([t_1, t_2]) = \{x \mid \exists x_0 \in X_0, t \in [t_1, t_2], x = x(t)\}.$$

For $\text{Reach}([0, t])$ we simply write $\text{Reach}(t)$.

Since exact computation of Reach is often intractable, one often resorts to using its over-approximations. Next, we provide a definition of *incremental practical stability* [3].

Definition 1. A system is *incrementally practically stable* if there exists a class- $\mathcal{KL}$ function β and a constant $c \geq 0$ such that, for any pair of initial states $x^{(1)}, x^{(2)} \in X_0$, the distance between the corresponding executions $\delta x(t) := x^{(1)}(t) - x^{(2)}(t)$ satisfies

$$\|\delta x(t)\| \leq \beta(\|\delta x(0)\|, t) + c, \quad \forall t \geq 0.$$

Recall that a $\mathcal{KL}$ function is strictly decreasing in its second argument and satisfies $\beta(r, t) \rightarrow 0$ as $t \rightarrow \infty$ for any fixed r . Therefore, the first term on the right-hand side of the bound in Definition 1 decays to zero over time, while the second term c provides a uniform bound on the distance between executions. The stronger property corresponding to $c = 0$ is called *incremental stability*. Incremental stability and existence of an equilibrium together imply global asymptotic stability. Incremental stability is an important requirement and has been used for developing data-driven verification algorithms [11, 12]. Unlike safety and stability properties, incremental stability involves quantification over multiple executions and from the standpoint of formal languages it is a hyperproperty [5].

Next, we present Theorem 1, which provides an upper bound on the deviation between executions of system (1), thereby offering a systematic method for establishing incremental practical stability. A key feature of this result is the use of the P -norm, which weights state deviations and captures anisotropic contraction behavior across different dimensions. By concretizing general class- $\mathcal{KL}$ functions in Definition 1 with exponential functions, Theorem 1 retains desirable properties for formalizing contraction while providing a flexible and broadly applicable structure for optimization.

Theorem 1. Let $\mathcal{R} \supseteq \text{Reach}([0, T])$ be a compact over-approximation of the reachable set for some $T > 0$, and let $\mathcal{J}_{\mathcal{R}} = \{J_f(x) \in \mathbb{R}^{n \times n} \mid x \in \mathcal{R}, f \in \mathcal{F}\}$ be the Jacobian set over $\mathcal{R}$. If $\mathcal{J}_{\mathcal{R}}$ is compact and convex and if there exists $P \in \mathbb{S}_{>0}^n$ and $\gamma \in \mathbb{R}$ such that $A^\top P + PA \prec \gamma P$ for all $A \in \mathcal{J}_{\mathcal{R}}$, then for any initial states $x^{(1)}, x^{(2)} \in X_0$, and any $\epsilon > 0$, there exists a constant $C_{P, \mathcal{R}} \geq 0$ such that for all $t \in [0, T]$:

$$\|\delta x(t)\|_P^2 \leq e^{\lambda t} \|\delta x(0)\|_P^2 + \frac{e^{\lambda t} - 1}{\lambda \epsilon} C_{P, \mathcal{R}}, \text{ where } \lambda = \gamma + \epsilon.$$

PROOF. Let $\delta x(t) := x^{(1)}(t) - x^{(2)}(t)$. From system (1), there exist selections $f_1, f_2 \in \mathcal{F}$ such that

$$\begin{aligned} \dot{\delta x} &= f_1(x^{(1)}) - f_2(x^{(2)}) = f_1(x^{(1)}) - f_1(x^{(2)}) + f_1(x^{(2)}) - f_2(x^{(2)}) \\ &= A \delta x + w, \text{ where } w = f_1(x^{(2)}) - f_2(x^{(2)}) \text{ and } A = \int_0^1 J_{f_1}(x^{(2)} + s(x^{(1)} - x^{(2)})) ds \end{aligned}$$

and the last step follows from the mean value theorem. Computing $\frac{d}{dt} \|\delta x\|_P^2$:

$$\begin{aligned} \frac{d}{dt} \|\delta x\|_P^2 &= \dot{\delta x}^\top P \delta x + \delta x^\top P \dot{\delta x} \\ &= \delta x^\top (A^\top P + PA) \delta x + w^\top P \delta x + \delta x^\top P w. \end{aligned}$$

Applying Proposition 3 with $\epsilon > 0$ bounds the cross term:

$$w^\top P \delta x + \delta x^\top P w = 2 \delta x^\top P w \leq \epsilon \delta x^\top P \delta x + \frac{1}{\epsilon} w^\top P w,$$

so that $\frac{d}{dt} \|\delta x\|_P^2 \leq \delta x^\top (A^\top P + PA) \delta x + \epsilon \delta x^\top P \delta x + \frac{1}{\epsilon} w^\top P w$. By Proposition 1, $A \in \mathcal{J}_R$, so $A^\top P + PA \prec \gamma P$ gives $\delta x^\top (A^\top P + PA) \delta x \leq \gamma \delta x^\top P \delta x$. By Proposition 2, $w^\top P w \leq C_{P,R}$. Substituting yields:

$$\frac{d}{dt} \|\delta x\|_P^2 \leq (\gamma + \epsilon) \|\delta x\|_P^2 + \frac{1}{\epsilon} C_{P,R}.$$

Setting $\lambda = \gamma + \epsilon$ and applying Grönwall's inequality gives:

$$\|\delta x(t)\|_P^2 \leq e^{\lambda t} \|\delta x(0)\|_P^2 + \frac{e^{\lambda t} - 1}{\lambda \epsilon} C_{P,R},$$

and replacing $\delta x(t)$ with $x^{(1)}(t) - x^{(2)}(t)$ concludes the proof. $\square$

Theorem 1 bounds the difference between any two executions of the set-valued system (1) over the finite time horizon T , based on initial conditions through the system via the P -norm and the constants γ , ϵ , and $C_{P,R}$.

We note that λ and $C_{P,R}$ depend on the set $\mathcal{J}_R$, which in turn depends on the reachable set $\mathcal{R}$ up to time horizon T . If we can over-approximate the unbounded time reach set $\text{Reach}(\infty)$, for example, via fixed-point checks, then the bound on $\delta x(t)$ can be extended arbitrarily.

A key factor in the asymptotic behavior of the deviation dynamics is $\lambda = \gamma + \epsilon$: if $\lambda > 0$, deviations grow over time; if $\lambda < 0$, deviations shrink, eventually remaining within a bounded tube, named the *ultimately invariant tube*, as formalized in the following corollary.

Corollary 1. *Under the assumptions of Theorem 1, if $\lambda < 0$ then system (1) is incrementally practically stable with*

$$\limsup_{t \rightarrow \infty} \|\delta x(t)\|_P^2 \leq \frac{-1}{\lambda \epsilon} C_{P,R}.$$

PROOF. From Theorem 1, for all $t \geq 0$:

$$\|\delta x(t)\|_P^2 \leq e^{\lambda t} \|\delta x(0)\|_P^2 + \frac{e^{\lambda t} - 1}{\lambda \epsilon} C_{P,R}.$$

Since $\lambda < 0$, $e^{\lambda t} \|\delta x(0)\|_P^2 \rightarrow 0$ and $\frac{e^{\lambda t} - 1}{\lambda \epsilon} C_{P,R} \rightarrow \frac{-1}{\lambda \epsilon} C_{P,R}$ as $t \rightarrow \infty$. Taking the limit superior gives the stated bound, which is finite and positive since $\lambda < 0$ and $\epsilon > 0$, confirming incremental practical stability per Definition 1. $\square$

3.3 Computing Incremental Convergence Rates

In this section, we formulate an optimization problem to compute bounds on the convergence rates. As established in Theorem 1, both γ and $C_{P,R}$ determine the tightness of the ultimately invariant tube: a smaller (more negative) γ accelerates contraction, while a smaller $C_{P,R}$ yields a tighter ultimate bound. Accordingly, our objective is to minimize both quantities.

To upper-bound $C_{P,R}$, we parameterize the difference between any two smooth selections by constructing a new mapping $(x, s) \mapsto \phi_s(x)$. For example, if $F(x) = \{kx + c \mid k \in [k_l, k_u], c \in [c_l, c_u]\}$, we define $\phi_{(s_1, s_2)}(x) =$

$s_1(k_u - k_l)x + s_2(c_u - c_l)$, which ensures $\{f_1(x) - f_2(x) \mid f_1, f_2 \in \mathcal{F}, x \in X\} = \{\phi_s(x) \mid x \in X, s \in [0, 1]^2\}$. We then study the function $(x, s) \mapsto \|\phi_s(x)\|_P$. By applying bound-propagation techniques (e.g., CROWN), we derive a linear relaxation of this mapping and obtain a constant upper bound on its output, yielding a feasible and tight estimate of $C_{P, \mathcal{R}}$.

For γ , Theorem 1 requires γ satisfying the Lyapunov-type matrix inequality $A^\top P + PA \prec \gamma P$ for all $A \in \mathcal{J}_{\mathcal{R}}$. Since $A^\top P + PA \preceq \lambda_{\max}(A^\top P + PA) I$ and $P \succeq \lambda_{\min}(P) I$, this condition is always satisfiable by selecting

$$\gamma \geq \max_{A \in \mathcal{J}_{\mathcal{R}}} \frac{\lambda_{\max}(A^\top P + PA)}{\lambda_{\min}(P)}.$$

Because a smaller (more negative) γ leads to a tighter contraction rate, we pose the search for the optimal contraction rate γ and corresponding metric P as the following optimization problem:

$$\begin{aligned} \min_{P \in \mathbb{S}_{>0}^n, \gamma \in \mathbb{R}} \quad & \gamma \\ \text{s.t.} \quad & A^\top P + PA \prec \gamma P, \quad \forall A \in \mathcal{J}_{\mathcal{R}}. \end{aligned} \tag{2}$$

For any fixed value of γ , the search for a feasible matrix $P \succ 0$ satisfying $A^\top P + PA \prec \gamma P$ for all $A \in \mathcal{J}_{\mathcal{R}}$ reduces to a semidefinite program (SDP). To obtain a suboptimal contraction rate, we perform a linear (or bisection) search over decreasing values of γ , checking feasibility of the SDP at each step; using a finer search resolution allows the resulting estimate of γ to approach the true optimal value arbitrarily closely. Moreover, when the Jacobian set $\mathcal{J}_{\mathcal{R}}$ is not only convex but also a convex polytope, the infinitely many constraints need only be enforced at the vertices of $\mathcal{J}_{\mathcal{R}}$, yielding a finite SDP representation.

Example: Position Control with State Feedback To illustrate incremental stability without any perception uncertainty for now, consider a one-dimensional system: $\dot{x} = -Kx$, where K is the control gain. The smooth cover for this set-valued system is $\mathcal{F} = \{-Kx\}$, and the Jacobian set is $\mathcal{J} = \{-K\}$, which is constant. With $P = 1$, the point-wise diameter is $C_P = 0$ since there is only one selection. For $K = 1$: The Jacobian is $A = -K = -1$. From Theorem 1, we need $A^\top P + PA \prec \gamma P$. With $P = 1$, this gives $(-1)(1) + (1)(-1) = -2 \prec \gamma$, so $\gamma > -2$. Let $\gamma = -1.9$. Choosing $\epsilon = 0.1$ gives $\lambda = \gamma + \epsilon = -1.8 < 0$. From Theorem 1, for any two initial states $x^{(1)}(0)$ and $x^{(2)}(0)$:

$$\|\delta x(t)\|_P^2 \leq e^{-1.8t} \|\delta x(0)\|_P^2 + \frac{e^{-1.8t} - 1}{-1.8 \cdot 0.1} \cdot 0 = e^{-1.8t} \|\delta x(0)\|_P^2.$$

Since $C_P = 0$, the distance between any two trajectories converges exponentially to zero, certifying incremental stability.

For $K = 2$: We have $A = -2$, giving $\gamma > -4$. Taking $\gamma = -3.9$ and $\epsilon = 0.1$ yields $\lambda = -3.8$, resulting in faster exponential convergence, as the larger gain K drives a more negative contraction rate λ : $\|\delta x(t)\|_P^2 \leq e^{-3.8t} \|\delta x(0)\|_P^2$.

3.4 From Incremental Stability to Asymptotic Stability

Having established bounds on incremental convergence rates in §3.3, we now leverage these results to certify asymptotic stability with respect to a target set $X^* \subseteq X$ via Lyapunov functions.

A dynamical system is *asymptotically stable with respect to a set* $X^* \subseteq X$ if $\forall x_0 \in X$, the execution $x(t)$ satisfies $\lim_{t \rightarrow \infty} d_{X^*}(x(t)) = 0$; this definition extends analogously to set-valued dynamical systems. A Lyapunov function is a standard certificate for proving asymptotic stability of dynamical systems. Here we define Lyapunov functions for set-valued continuous-time dynamical systems and provide a sufficient condition for asymptotic stability of such systems.

Theorem 2 (Lyapunov Condition for Asymptotic Stability). *For system (1), given 1) a nonempty, compact set $X^* \subset X$; 2) a continuously differentiable function $V : X \rightarrow \mathbb{R}_{\geq 0}$ such that $V(x) > c$ for all $x \in X \setminus X^*$ and $V(x) \leq c$ for all $x \in X^*$; and 3) a continuous function $W : X \rightarrow \mathbb{R}_{\geq 0}$ such that $W(x) = 0$ iff $x \in X^*$. If*

$$\nabla V(x) \cdot f(x) \leq -W(x), \quad \forall x \in X, f \in \mathcal{F},$$

then X^ is asymptotically stable. V is called a Lyapunov function for the system.*

While Theorem 2 certifies convergence for each individual execution, it does not account for the nondeterminism introduced by the set-valued dynamics F . The incremental stability framework developed in §3.2–3.3 bridges this gap: once a reference execution is known to converge to X^* , the incremental stability bound transfers this convergence to all executions. We now formalize this connection.

Theorem 3 (Incremental Stability implies Asymptotic Stability). *Consider system (1) with smooth cover $\mathcal{F}$, and let $X^* \subset X$ be a nonempty compact set. Suppose that (1) the conditions of Theorem 1 hold with $\lambda = \gamma + \epsilon < 0$; (2) there exists a reference execution $x^*(t)$ such that $d_{P,X^*}(x^*(t)) \rightarrow 0$ as $t \rightarrow \infty$; (3) the point-wise P -diameter satisfies*

$$\frac{-1}{\lambda\epsilon} C_{P,\mathcal{R}} \leq \mu.$$

Then system (1) is asymptotically stable with respect to X^ .*

Equivalently, condition (3) requires that there exists $\mu > 0$ such that $C_{P,\mathcal{R}} \leq -\lambda\epsilon\mu$.

PROOF. We verify the Lyapunov condition from Theorem 2 using:

$$V(\delta x) = \|\delta x\|_P^2, \quad W(\delta x) = -\lambda \cdot \text{ReLU}(\|\delta x\|_P^2 - \mu).$$

The gradient of V is $\nabla V(\delta x) = 2\delta x^\top P$. From Theorem 1, the dynamics of δx can be written as $f(\delta x) = A\delta x + w$. Therefore:

$$\begin{aligned} \nabla V \cdot f(\delta x) &= 2\delta x^\top P(A\delta x + w) = \delta x^\top (PA + A^\top P)\delta x + 2\delta x^\top Pw \\ &\leq \lambda \|\delta x\|_P^2 + \frac{1}{\epsilon} C_{P,\mathcal{R}}. \end{aligned}$$

By condition (3), $C_{P,\mathcal{R}} \leq -\lambda\epsilon\mu$, so:

$$\nabla V \cdot f(\delta x) \leq \lambda \|\delta x\|_P^2 - \lambda\mu = \lambda(\|\delta x\|_P^2 - \mu).$$

For $\delta x \notin X^*$ (i.e., $\|\delta x\|_P^2 > \mu$), we have $\|\delta x\|_P^2 - \mu > 0$, so:

$$\lambda(\|\delta x\|_P^2 - \mu) = \lambda \cdot \text{ReLU}(\|\delta x\|_P^2 - \mu) = -W(\delta x).$$

Thus $\nabla V \cdot f(\delta x) \leq -W(\delta x)$ for all $\delta x \notin X^*$, establishing asymptotic stability by Theorem 2. $\square$

REMARK 1. *In the limiting case $C_{P,\mathcal{R}} = 0$, the set-valued dynamics reduces to a single-valued map, and incremental stability with $\lambda < 0$ directly implies asymptotic stability, recovering the classical result that incremental stability together with existence of an equilibrium implies asymptotic stability.*

Example Continued: Asymptotic Stability Recall the position control system from Section 3.3. Setting $\dot{x} = 0$ gives the equilibrium at $x^* = 0$. We now verify asymptotic stability with respect to $X^* = \{0\}$. Directly using Theorem 2,

consider the Lyapunov function $V(x) = x^2$ with $P = 1$. For the dynamics $f(x) = -Kx$:

$$\nabla V \cdot f(x) = 2x \cdot (-Kx) = -2Kx^2.$$

Setting $W(x) = 2Kx^2$, we have $W(x) = 0$ if and only if $x = 0$, and $\nabla V \cdot f(x) = -2Kx^2 \leq -W(x)$ for all x . By Theorem 2, $X^* = \{0\}$ is asymptotically stable.

Using incremental stability via Theorem 3: In Section 3.3, we established incremental stability with $\lambda = -1.8 < 0$ and $C_P = 0$. Since $C_{P,R} = 0$, condition (3) is satisfied for any $\mu > 0$:

$$\frac{-1}{\lambda\epsilon} C_{P,R} = 0 \leq \mu.$$

By Theorem 3, asymptotic stability follows from incremental stability; with $C_{P,R} = 0$, incremental stability directly implies asymptotic stability.

4 Perception Contracts for Vision-Based Control

4.1 Concrete and Abstract Systems

The state evolution of the system is given by

$$\begin{aligned} \dot{x} &= f(x) + g(y), \quad x(0) \in X_0 \\ y &= h(x, e), \quad e \in E, \end{aligned} \tag{3}$$

where $f : X \rightarrow X$ is the system dynamics, $g : Y \rightarrow X$ is a controller, and $y \in Y$ is the observation or state estimate generated by the perception module h . The observation y generated from state x depends on environmental factors e , such as lighting, weather, and environment geometry, and E is the set of all values that e can take.

A *perception contract* for h over E is an *environment-independent* set-valued function $M_E : X \rightarrow 2^Y$ such that $h(x, e) \in M_E(x)$ for all $e \in E$. Additionally, we require that M_E admits a smooth cover $\mathcal{M}_E$ —a family of smooth functions such that $M_E(x) = \{m(x) \mid m \in \mathcal{M}_E\}$ for all $x \in X$.

Using this contract, we define an *abstract system* that over-approximates the concrete system (3):

$$\begin{aligned} \dot{x} &= f(x) + g(y), \quad x(0) \in X_0, \\ y &\in M_E(x). \end{aligned} \tag{4}$$

Expanding the abstract system yields the set-valued dynamics:

$$\dot{x} \in F(x), \quad F(x) = \{f(x) + g(y) \mid y \in M_E(x)\}. \tag{5}$$

This is precisely the continuous-time analogue of the form (1) studied in Section 3.2, with smooth cover $\mathcal{F} = \{f + g \circ m \mid m \in \mathcal{M}_E\}$. The relationship between the concrete and abstract systems is illustrated in Figure 1. The concrete system has environment-dependent observations $h(x, e)$, while the abstract system replaces these with the set-valued contract $M_E(x)$.

The following proposition establishes that incremental stability of the abstract system guarantees incremental stability of the concrete system.

Proposition 4. *If the abstract system (4) is incrementally practically stable, then so is the concrete system (3).*

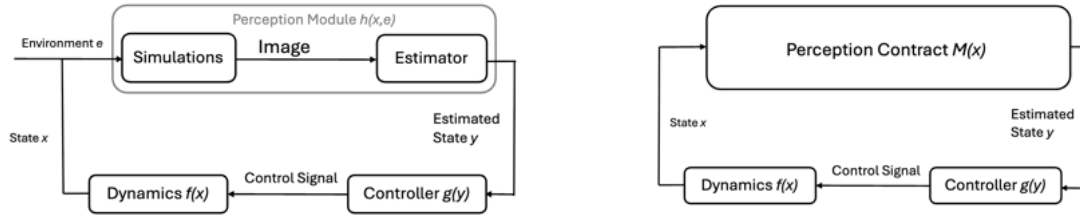


Fig. 1. Pipeline of Concrete System (left) and Abstract System (right). The concrete system has environment-dependent perception $h(x, e)$, while the abstract system uses the environment-independent contract $M_E(x)$.

PROOF. For any state $x(t) \in X$ and environmental factor $e \in E$, since $h(x, e) \in M_E(x)$ and M_E admits a smooth cover $\mathcal{M}_E$, there exists a smooth function $m_t \in \mathcal{M}_E$ such that

$$h(x(t), e) = m_t(x(t)).$$

Here, $x^c(t)$ and $x^a(t)$ denote trajectories of the concrete and abstract systems, respectively. Thus, for any trajectory $x^c(t)$ of system (3) with respect to a specific environmental factor e , we can associate a time-varying smooth function $m_t \in \mathcal{M}_E$ satisfying

$$\dot{x}^c = f(x^c) + g(h(x^c, e)) = f(x^c) + g(m_t(x^c)).$$

Hence every trajectory of system (3) corresponds to a trajectory of system (4). Consider any two trajectories starting from initial states $x^{c(1)}(0)$ and $x^{c(2)}(0)$ of system (3). The distance between these trajectories

$$\delta x^c(t) := x^{c(1)}(t) - x^{c(2)}(t)$$

equals the distance between the corresponding two trajectories of system (4), denoted $\delta x^a(t)$. Since system (4) is incrementally practically stable, there exist a class- $\mathcal{KL}$ function β and constant $c \geq 0$ such that

$$\|\delta x^c(t)\| = \|\delta x^a(t)\| \leq \beta(\|\delta x^a(0)\|, t) + c = \beta(\|\delta x^c(0)\|, t) + c.$$

Therefore, system (3) is also incrementally practically stable. $\square$

Proposition 4 allows the analysis of incremental stability for a concrete system subject to unformalized perception-based noise via an abstract system with a formally defined uncertainty bound.

4.2 Running Example: Adding Vision-Based Perception

Recall the 1D position control system from Sections 3.3 and 3.4, where we analyzed the perfect state feedback case $\dot{x} = -Kx$ with $C_P = 0$. We now introduce vision-based perception to illustrate how environmental uncertainty degrades closed-loop performance.

The controller now relies on a camera-based position estimate $y = h(x, e)$ instead of perfect state feedback, giving dynamics:

$$\dot{x} = -Ky, \tag{6}$$

where the observer output depends on environmental lighting conditions $e \in E = [0, 1]$ ($0 = \text{dark}$, $1 = \text{bright}$). The perception error is $\eta(e) = 0.1(1 - e)$, yielding the perception contract:

$$M_E(x) = [x - 0.1, x + 0.1]. \tag{7}$$

The abstract system is $\dot{x} \in F(x)$ where $F(x) = \{-Ky \mid y \in [x - 0.1, x + 0.1]\}$. The smooth cover is $\mathcal{F} = \{-K(x + \delta) \mid \delta \in [-0.1, 0.1]\}$. The Jacobian set remains $\mathcal{J} = \{-K\}$, but now the point-wise diameter is $C_P = 0.2K > 0$ (with $P = 1$).

For $K = 1$: We have $\gamma > -2$ (unchanged), taking $\gamma = -1.9$ and $\epsilon = 0.1$ gives $\lambda = -1.8 < 0$. From Corollary 1:

$$\limsup_{t \rightarrow \infty} \|\delta x(t)\|_P^2 \leq \frac{0.2}{(-1.8)(0.1)} = \frac{0.2}{0.18} \approx 1.11, \quad \text{or} \quad \|\delta x(t)\|_P \leq \sqrt{1.11} \approx 1.05.$$

For $K = 2$: With $\gamma = -3.9$, $\epsilon = 0.1$, $\lambda = -3.8$, and $C_P = 0.4$, the bound is $\|\delta x(t)\|_P^2 \leq \frac{0.4}{0.38} \approx 1.05$, or $\|\delta x(t)\|_P \leq \sqrt{1.05} \approx 1.03$.

Impact of perception uncertainty: Without perception (Section 3.3), trajectories converged exponentially to zero with $C_P = 0$. With the perception contract, $C_P = 0.2K > 0$ creates an ultimately invariant tube—trajectories no longer reach zero but remain bounded within radius ≈ 1.05 for $K = 1$. This quantifies the performance degradation induced by vision-based estimation under environmental variation.

4.3 Learning Perception Contracts from Data

We now present an algorithm for constructing perception contracts from sampled data. The key idea is to estimate both a center function $M_c(x)$ that approximates the nominal observer output and a radius function $M_r(x)$ that captures the deviation under environmental variations.

Algorithm 1 takes as input the target state and environment regions (X, E) , the conformance measure pr , error tolerance ϵ , and confidence parameter δ . The algorithm first determines the number of samples N required to achieve the desired statistical conformance (Line 1). It then samples pairs of states and environmental parameters (x_i, e_i) from a predefined distribution $\mathcal{D}$ and computes the corresponding perception module outputs $y_i = h(x_i, e_i)$ (Lines 2–3). Next, it estimates the center function $M_c(x)$ of the perception contract via regression using the collected data $\{(x_i, y_i)\}$ (Line 4). It then computes the radius function $M_r(x)$ through quantile regression [9], ensuring that at least a proportion pr of samples satisfy $|y_i - M_c(x_i)| \leq M_r(x_i)$ (Line 5). The outputs, M_c and M_r , together define the learned perception contract M_E , which is a set-valued mapping such that for all $x \in X$:

$$M_E(x) = \{y \mid \|y - M_c(x)\| \leq M_r(x)\}. \quad (8)$$

The following proposition establishes that the learned contract satisfies the target conformance measure with high probability.

Proposition 5. *With probability at least $1 - \delta$, the actual conformance measure*

$$p = \mathbb{E}_{(x,e) \sim \mathcal{D}} [\mathbb{I}(h(x,e) \in M_E(x))] \quad (9)$$

of the learned perception contract M_E is at least pr , as defined in Algorithm 1.

PROOF. From Hoeffding's inequality, we have:

$$\Pr(p - \hat{p} \geq \epsilon) \leq e^{-2N\epsilon^2} = e^{-2N \cdot (-\frac{\ln \delta}{2N})} = \delta.$$

This implies

$$\Pr(\hat{p} - p \leq \epsilon) \geq 1 - \delta$$

completing the proof. $\square$

Algorithm 1 LearnCtrt: Learning Perception Contract from Data**Input:** State space X , environment E , perception h , conformance pr , tolerance ϵ , confidence δ **Output:** Center function M_c , radius function M_r

```

1:  $N \leftarrow -\frac{\ln \delta}{2\epsilon^2}$ 
2:  $\{(x_i, e_i)\}_{i=1}^N \leftarrow \text{SAMPLE}_{\mathcal{D}}(X, E, N)$ 
3:  $\{y_i\}_{i=1}^N \leftarrow \{h(x_i, e_i)\}_{i=1}^N$ 
4:  $M_c \leftarrow \text{REGRESSION}(\{(x_i, y_i)\}_{i=1}^N)$ 
5:  $M_r \leftarrow \text{QUANTILEREGRESSION}(\{(x_i, \|y_i - M_c(x_i)\|)\}_{i=1}^N, pr + \epsilon)$ 
6: return  $\langle M_c, M_r \rangle$ 

```

Proposition 5 provides a probabilistic bound on the observation outputs under environmental variations E . Its validity follows from the Hoeffding inequality, which ensures that the empirical bounds $M_E(x)$ estimated from samples generalize to the true distribution with high confidence $1 - \delta$. Incorporating the perception contract as an over-approximation of perception module h into system (3) yields the abstract system (4).

4.4 Incremental Stability Guided Refinement of Perception Contracts

In this section, we introduce an iterative refinement algorithm that identifies an environmental region $\tilde{E} \subseteq E$ for which a synthesized perception contract guarantees incremental stability of the resulting set-valued closed-loop system, and we establish the correctness of this procedure.

Algorithm 2 Identifying Environmental Regions That Guarantee Incremental Stability under Perception Contracts**Input:** $X, X_0, E, T, pr, \epsilon, \delta, \epsilon, Thr$ **Output:** $\langle \tilde{E}, M_{\tilde{E}}, P, \lambda, C_{P, \mathcal{R}} \rangle$

```

1:  $\tilde{E} \leftarrow E$ 
2: repeat
3:    $M_{\tilde{E}} \leftarrow \text{LPC}(X, \tilde{E}, pr, \delta, \epsilon)$ 
4:    $\mathcal{F} \leftarrow \{f + g \circ m \mid m \in M_{\tilde{E}}\}$ 
5:    $\mathcal{R} \leftarrow \text{Over-Approx}(\mathcal{F}, X_0, T)$ 
6:    $\mathcal{J}_{\mathcal{R}} \leftarrow \text{Jacobians}(\mathcal{R})$ 
7:    $(\gamma, P) \leftarrow \text{solve}(\text{OPT}(\mathcal{J}_{\mathcal{R}}))$ 
8:    $\lambda \leftarrow \gamma + \epsilon$ 
9:    $C_{P, \mathcal{R}} \leftarrow \sup_{f_1, f_2 \in \mathcal{F}, x \in \mathcal{R}} \|f_1(x) - f_2(x)\|_P^2$ 
10:   $\tilde{E} \leftarrow \text{shrink}(\tilde{E})$  ▷ applied only if condition not met
11: until  $e^{\lambda T} d_P^2(X_0) \leq Thr$  and  $\lambda < 0$ 
12: return  $\langle \tilde{E}, M_{\tilde{E}}, P, \lambda, C_{P, \mathcal{R}} \rangle$ 

```

Algorithm 2 takes as input the state space X , initial set X_0 , initial environment E , time horizon T , perception-contract parameters (pr, ϵ, δ) , scalar $\epsilon > 0$, and threshold $Thr > 0$, and returns the refined region $\tilde{E}$, its corresponding perception contract $M_{\tilde{E}}$, norm matrix P , contraction rate λ , and the uniform bound $C_{P, \mathcal{R}}$. The procedure begins by initializing $\tilde{E}$ (Line 1), computing a perception contract over this region (Line 3), and constructing the associated smooth cover $\mathcal{F}$ of closed-loop dynamics (Line 4). It then over-approximates the reachable set over the horizon T (Line 5) and extracts the Jacobians attainable along this set (Line 6). Using these Jacobians, the contraction optimization problem (2) is solved to obtain (γ, P) via SDP solver (Line 7), from which the contraction factor $\lambda = \gamma + \epsilon$ and the uniform deviation bound $C_{P, \mathcal{R}}$ are computed (Lines 8–9). If the predicted deviation after time T satisfies the threshold condition $e^{\lambda T} d_P^2(X_0) \leq Thr$, the

loop terminates and the algorithm returns; otherwise, the environment region $\tilde{E}$ is iteratively reduced and re-evaluated until stability is certified (Line 10).

Theorem 4. *If Algorithm 2 terminates and returns $\langle \tilde{E}, M_{\tilde{E}}, P, \lambda, C_{P,\mathcal{R}} \rangle$, then for the abstract closed-loop system over $\tilde{E}$, the deviation $\delta x(t)$ between any two executions satisfies*

$$\|\delta x(T)\|_P^2 \leq Thr + \frac{e^{\lambda T} - 1}{\lambda \epsilon} C_{P,\mathcal{R}}.$$

PROOF. From Line 4, the abstract closed-loop system over $M_{\tilde{E}}$ has smooth cover $\mathcal{F}$. Applying Theorem 1 with $\mathcal{R}, \mathcal{J}_{\mathcal{R}}, \gamma, P, \lambda, C_{P,\mathcal{R}}$ from Lines 5–9 gives

$$\|\delta x(t)\|_P^2 \leq e^{\lambda t} \|\delta x(0)\|_P^2 + \frac{e^{\lambda t} - 1}{\lambda \epsilon} C_{P,\mathcal{R}}.$$

Using $\|\delta x(0)\|_P^2 \leq d_P^2(X_0)$ and $e^{\lambda T} d_P^2(X_0) \leq Thr$, and evaluating at $t = T$ yields the conclusion. $\square$

As $T \rightarrow \infty$, since $\lambda < 0$ the first term vanishes and the bound is dominated by the steady term $\frac{1}{\lambda \epsilon} C_{P,\mathcal{R}}$, showing that the system is incrementally stable.

4.5 Lyapunov Perception Contracts

To identify the subset $E_0 \subseteq E$ of environments in which system (3) is asymptotically stable with respect to X^* , we introduce the notion of a *Lyapunov perception contract*. The key idea is to construct an environment-independent over-approximation $M : X \rightarrow 2^Y$ of the observer h such that the resulting set-valued system

$$\dot{x} \in F(x), \quad F(x) = \{f(x) + g(y) \mid y \in M(x)\}, \quad (10)$$

which is precisely of the form (5) studied in Section 3.2, admits a Lyapunov function V certifying asymptotic stability with respect to X^* . The output of the observer $h(x, e)$ can be made to be contained in M by simply making M large or conservative. However, such an over-approximation may not be useful for establishing system stability. Thus, M has to balance observer conformance and system stability, which leads to the following definition.

Definition 2. For system (3), given a target set $X^* \subset X$ and a candidate Lyapunov function V , a mapping $M : X \rightarrow 2^Y$ is called a $\langle E, X^* \rangle$ -Lyapunov perception contract if it satisfies:

- (1) *Conformance:* $\forall x \in X, \forall e \in E, h(x, e) \in M(x)$;
- (2) *Correctness:* V is a Lyapunov function with respect to X^* for system (10).

Proposition 6. *For system (3), if M is a $\langle E, X^* \rangle$ -Lyapunov perception contract, then system (3) is asymptotically stable with respect to X^* .*

Proposition 6 follows from the fact that $f(x) + g(h(x, e)) \in F(x)$ over X and E . This implies that if V is a Lyapunov function for the set-valued system (10), then it is also a Lyapunov function for system (3). Asymptotic stability of (3) then follows from Theorem 2.

Regarding the construction of the Lyapunov function, we note that the controller for a visual autonomous system is usually designed first with an ideal observer that has access to ground truth values. This ideal observer is then approximately implemented using the perception pipeline. The Lyapunov function for that ideal controller can serve as an initial guess for V , and the perception contract M is constructed from sampled data using LEARNCTR (Algorithm 1), as described in Section 4.3.

Lyapunov perception contracts represent a specialization of the incremental stability framework developed in §3.2–3.3 to systems that admit a target set X^* and a Lyapunov function certifying convergence to it. Recall from Theorem 3 that incremental stability with $\lambda < 0$ implies asymptotic stability with respect to X^* , provided a reference execution converges to X^* and the tube radius $\frac{1}{\lambda\epsilon}C_{P,\mathcal{R}}$ fits inside X^* . An LPC directly supplies both ingredients: the correctness condition in Definition 2 certifies via V that every execution of system (10) converges to X^* , while the conformance condition ensures M faithfully over-approximates h across all $e \in E$.

4.6 ODD Identification via Requirement-Guided Refinement

A key application of Lyapunov perception contracts is identifying the *operating design domain* (ODD), the largest environment subset $\tilde{E} \subseteq E$ under which asymptotic stability of system (3) can be formally certified. For certain environments $e \in E$, the system may not be stable, and therefore even a conformant contract will not establish correctness. The idea is to iteratively shrink the environment over which the contract is constructed until the Lyapunov correctness condition is satisfied. We now present the *Data and Requirement Guided Lyapunov Perception Contract* (DARELPC) algorithm (Algorithm 3), which uses LEARNCTRT, presented in Section 4.3, to construct a conformant perception contract, and applies CHECKLYAP to verify the Lyapunov correctness condition.

$\text{CHECKLYAP}(\tilde{X}, M_{\tilde{E}}, V, X^*)$. The function CHECKLYAP takes as input a set of states $\tilde{X}$, a candidate perception contract $M_{\tilde{E}}$, a candidate Lyapunov function V , and a target set X^* . The function first partitions $\tilde{X}$ into fine grids and computes the upper bound of the time derivative of V for each partition via the linear bound propagation tool CROWN [46]. The function then checks whether $\tilde{X} \setminus X^*$ intersects with the region where $\dot{V}$'s upper bound is non-negative. If no such intersection exists, it implies that all states of $\tilde{X} \setminus X^*$ have a strictly negative Lyapunov time derivative for system (10). According to Theorem 2, this means V is a valid Lyapunov function with respect to X^* . Otherwise, asymptotic stability is not guaranteed.

Verifying the Lyapunov condition over the entire state space is challenging, especially since the perception error, and thus the radius of the perception contract, can become too large for the condition to hold. Therefore, DARELPC takes as input the region of interest $\tilde{X} \subseteq X$, an environment set E_0 , the target set X^* , the parameters of LEARNCTRT, and the candidate Lyapunov function V . DARELPC confirms the correctness requirement of the computed Lyapunov perception contract by checking if V is a Lyapunov function of system (10) with respect to X^* using CHECKLYAP. If V is indeed a Lyapunov function with respect to X^* for system (10), then a Lyapunov perception contract $M_{\tilde{E}}$ is found and the corresponding environment range is returned. Otherwise, the algorithm shrinks the range of allowed environment $\tilde{E}$ via SHRINKENV. The function SHRINKENV takes a set of environmental parameters $\tilde{E}$ and generates a new, smaller set. The sequence of sets generated by repeatedly applying SHRINKENV monotonically converges to a singleton set containing only a nominal environment e_d . DARELPC is sound, in the sense that:

Theorem 5. *The output $M_{\tilde{E}}$ of Algorithm DARELPC is a $\langle \tilde{E}, X^* \rangle$ -LPC for system $\dot{x} = f(x) + g(o(x, e))$.*

Under the assumption that system (3) is asymptotically stable under nominal environment e_d , we are also able to show DARELPC is also complete.

4.7 Abstract Rendering for Perception Contracts

An alternative approach to constructing perception contracts is through *abstract rendering* [17], which computes provable bounds on all images rendered under continuously varying scene and background conditions. Rather than

Algorithm 3 DaReLPC: ODD Identification via Requirement-Guided Refinement

```

1: Function DARELPC( $\tilde{X}, E, e_d, X^*, o$ )
2:   params:  $pr, \epsilon, \delta, V$ 
3:    $\tilde{E} \leftarrow E$ 
4:    $M_{\tilde{E}} \leftarrow \text{LEARNCTRT}(\tilde{X}, \tilde{E}, o, pr, \epsilon, \delta)$ 
5:   while True do
6:     if CHECKLYAP( $\tilde{X}, M_{\tilde{E}}, V, X^*$ ) then
7:       break
8:      $\tilde{E} \leftarrow \text{SHRINKENV}(\tilde{E}, e_d)$ 
9:      $M_{\tilde{E}} \leftarrow \text{LEARNCTRT}(\tilde{X}, \tilde{E}, o, pr, \epsilon, \delta)$ 
10:  return  $\langle M_{\tilde{E}}, \tilde{E} \rangle$ 

```

learning a contract from sampled observations as in LEARNCTRT, abstract rendering derives the contract directly from the scene representation by propagating scene/background uncertainty through the rendering pipeline itself.

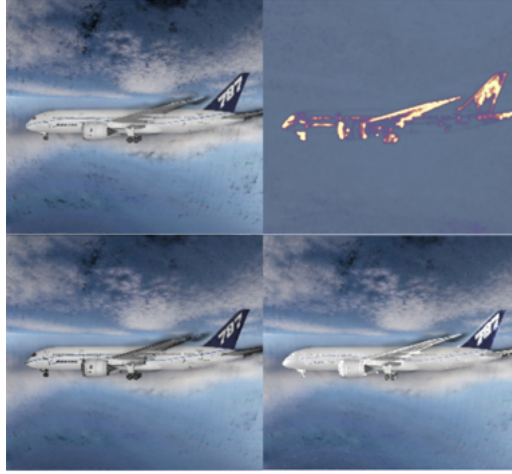


Fig. 2. Abstract rendering applied to a scene of an airplane in the sky. Top left: the rendered image of the airplane. Bottom left: the lower bound image. Bottom right: the upper bound image. Top right: a heat map showing the pixel-wise difference between the upper and lower bound images.

Given a 3D scene representation, abstract rendering computes lower and upper bounds $[\underline{\text{img}}(x), \overline{\text{img}}(x)]$ such that every image rendered under any background disturbances, or environmental condition $e \in E$ is guaranteed to lie within these bounds. The resulting *abstract image*, expressed as a set of constraints over the image matrix, is a sound over-approximation of all photometric variation induced by scene and background uncertainty. Composing this abstract image with a downstream observer h then yields a perception contract $M_E(x)$ that over-approximates the observer output for all $x \in X, e \in E$:

$$\underline{\text{img}}(x) \leq \text{img}(x, e) \leq \overline{\text{img}}(x), \quad (11)$$

and the perception contract is then defined as:

$$M_E(x) = \{h(\text{img}') \mid \text{img}' \in [\underline{\text{img}}(x), \overline{\text{img}}(x)]\}. \quad (12)$$

Abstract rendering targets two state-of-the-art photorealistic scene representations: 3D Gaussian Splatting [20] and Neural Radiance Fields (NeRF) [30]. The rendering pipeline for both representations is decomposed into compositions of

basic operations, and piecewise linear bounds are propagated through each step using CROWN [46]. Certain rendering-specific operations, including matrix inversion, sorting-based aggregation, and cumulative product summation, are not handled by standard neural network verification tools and require novel linear over-approximation techniques. The resulting framework is the first to deliver tight, end-to-end formal bounds for rendered images across continuous ranges of scene/background conditions, scaling to complex photorealistic scenes.

Compared to the sampling-based LEARNCTRT approach, abstract rendering provides tighter contracts when a 3D scene reconstruction is available, since it directly exploits the geometric structure of the scene rather than relying on statistical coverage. Contracts constructed via abstract rendering are expressed as linear sets (zonotopes) over pixel values rather than axis-aligned balls, capturing correlations between observer outputs that interval-based contracts cannot. This tightness carries through to the incremental stability analysis: tighter perception contracts yield smaller Jacobian sets $\mathcal{J}_{\mathcal{R}}$ and smaller point-wise diameters $C_{P,\mathcal{R}}$, directly reducing the ultimately invariant tube diameter certified by Theorem 1. A 3D scene representation is not required by the framework; abstract rendering is simply an alternative method for constructing perception contracts when one is available. However, when a 3D scene representation is not available, LEARNCTRT applies directly.

5 Experimental Evaluations

We validate the incremental stability framework on two vision-based control tasks: formation flying with 6D rigid-body aircraft dynamics using GateNet and UNet+PnP perception pipelines in photorealistic Unreal Engine environments, and automated landing (AutoLand) using Lyapunov perception contracts.

5.1 Formation Flying: Incremental Stability Certification

In this section, we present simulation-based experimental evaluation of our approach on a formation flying problem. The scenario setup is as follows: a *follower* aircraft pursues a *leader* using vision-based pose estimator. Specifically, the 6-dimensional relative pose (3D-translation and rotation defined by Euler angles) is estimated from the images from an onboard camera mounted on the leader using perception pipelines implemented using neural network models (see Figure 1). These relative pose estimates (y) are used by the follower to track the leader using its controller (g). We are interested in establishing incremental stability of the follower as it corresponds to its control performance in tracking the leader’s trajectory.

5.1.1 Overview of Configurations and Metrics. We consider two dynamic models (f) for the system: a 4D kinematic model [35] and a 6D rigid-body aircraft model [21]. Since controller synthesis is not the focus of this work, we consider only one state-feedback controller for each dynamics model: the standard nonlinear stable tracking (Kanayama) controller [19] for the 4D kinematic model, and a full-state feedback nonlinear controller [36] for the 6D rigid-body aircraft.

We use Unreal Engine for simulating the system and for generating photorealistic images under various environmental conditions, which feed different perception pipelines under investigation. The environmental variations (E) include changes in ambient light intensity and direction, fog density, and the background landscape that can take one of five values: island, seashore, valley, salt dome, and plains, as shown in Figure 4. We consider two different perception pipelines (h): a GateNet [34]-based estimator and a UNet+PnP-based estimator [28]. We will consider two types of perception contracts—linear and quadratic polynomial—with differing confidence levels to quantify perception uncertainty.

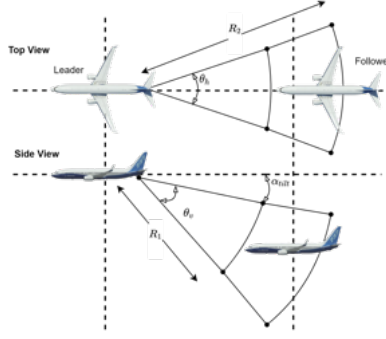


Fig. 3. Leader and follower aircraft and the camera's viewing frustum defined by the mounting angle (α_{ilt}), range R_1 to R_2 , and the horizontal (θ_h) and vertical (θ_v) fields of view.



Fig. 4. Unreal Engine images of follower aircraft with different variations in E: Normal Dome (top left), Dark Dome (top mid), Foggy Dome (top right), DirLight Dome (bottom left), Normal Island (bottom mid), Normal Seashore (bottom right).

For a given system configuration, the performance is measured using two metrics: The contraction rate λ and the diameter Dia of *Ultimately Invariant Set*, as computed by our method.

$$Dia = 2\sqrt{\frac{1}{-\lambda\epsilon}} \cdot \sqrt{\frac{C_{P,\mathcal{R}}}{\lambda_{\min}(P)}}.$$

Recall that $\lambda < 0$ certifies exponential incremental stability. We use Dia —rather than $C_{P,\mathcal{R}}$ —to measure the tightness of the ultimately invariant set, as it eliminates the scaling effect of $\det(P)$. Note that if $\lambda \geq 0$, the denominator in the computation of Dia becomes non-positive, rendering the resulting value meaningless.

5.1.2 Learning Perception Contracts. In this section, we describe the construction of perception contracts for two different perception pipelines using data generated from Unreal Engine simulations using Algorithm 1.

Perception pipelines (h). We are interested in evaluating two different perception pipelines for the formation flying application. The first uses GateNet [34]—a convolutional neural network for detecting the center and pose of a gate using only images from a single RGB camera for drone racing. For the formation flight application we trained two separate GateNet models for estimating the relative translation and the rotation. The second perception pipeline is

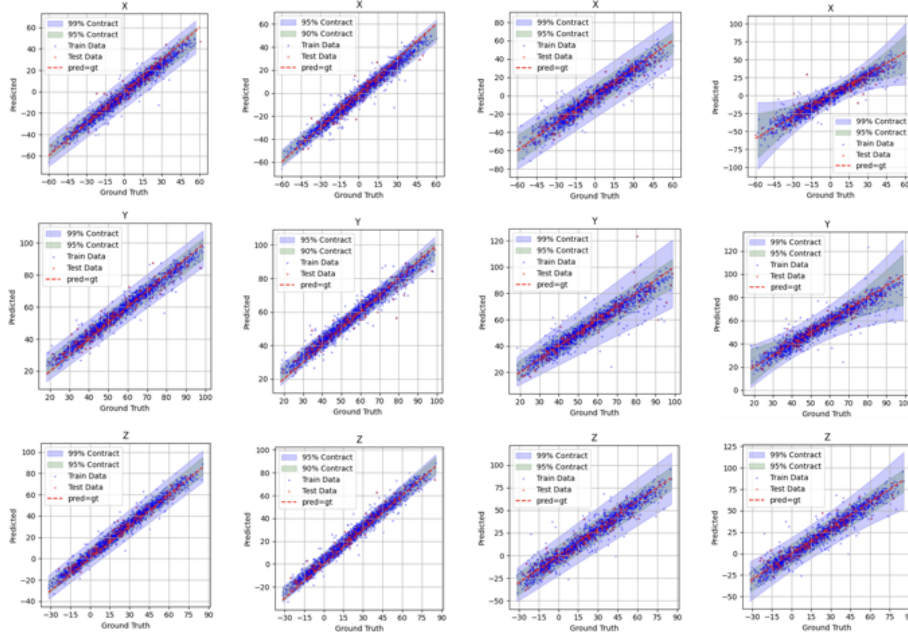


Fig. 5. Perception contracts across estimator and contract types. Rows correspond to translational states x (top), y (middle), and z (bottom). Columns correspond to: (1) Gatenet-linear, (2) Gatenet-quadratic polynomial, (3) UNet+PnP-linear, (4) UNet+PnP-quadratic polynomial. Each plot shows contracts under variations in lighting intensity, lighting direction, fog, and background. Red dashed line: perfect estimation ($\hat{x} = x$); blue/green regions: 99%/95% confidence bounds; blue/red points: training/test ground truth–estimate pairs.

based on the system in [28] and it uses a UNet [37] to detect a set of keypoints on the follower aircraft and then uses a Perspective-n-Point (PnP) algorithm [23] to compute the relative pose.

State (X) and environment (E) ranges. We assume that the follower aircraft is within the camera frustum (see Figure 3) defined by the leader’s camera¹: aircraft relative distance $R \in [40 \text{ m}, 120 \text{ m}]$, elevation angle $\theta_o \in [10^\circ, 40^\circ]$, and yaw angle $\theta_h \in [-30^\circ, 30^\circ]$. The data set over X in Algorithm 1 is generated by sampling uniformly over this set.

We consider the environmental variations (E) defined by the following parameter ranges: light intensity $[0.2, 1.0]$, sun light direction $[-45^\circ, 45^\circ]$ fog density $[0.0, 0.5]$, and five types of background scenes, as illustrated in Figure 4. The data set over E in Algorithm 1 is generated by sampling uniformly over this set E of environmental configurations.

Figure 5 shows different perception contracts $M(x)$ computed by Algorithm 1, with confidence parameter $pr = 0.95$ and $pr = 0.99$, for the GateNet pose estimator, under the above-mentioned state and environmental variations. Two sets of contracts are constructed with the linear and quadratic polynomial forms, and not surprisingly, that quadratic polynomial form contract $M(\cdot)$ is tighter in most parts of the state space. Similar trends are observed when perception contracts are constructed for the UNet+PnP estimator. As we shall see below, tighter contracts lead to less conservative abstract systems, and therefore, stronger incremental stability certificates.

5.1.3 Incremental Stability of Abstract Systems. We present the analysis of the abstract systems constructed from the perception contracts of Section 5.1.2. The abstract system (as in Equation (4)) is obtained by substituting the perception pipeline h (GateNet or UNet+PnP) with the contract M , in the concrete system. For each abstract system, we compute

¹Issues of losing the follower because it falls outside of this visibility cone are not relevant for this application.

the contraction rate λ and the diameter of the ultimately invariant set Dia . A smaller Dia indicates a tighter ultimately invariant set, while a smaller (more negative) λ corresponds to faster contraction. Throughout, the dynamics f and controller g are fixed as described in Section 5.1.1; controller design is not a contribution of this work. If the controller performs poorly, the system may diverge and the contraction analysis will fail to certify stability. This is the expected behavior, as our framework verifies but does not synthesize controllers.

Dyn (f)	Contract(M)	Estimator(h)	Env(E)	pr(%)	lead	λ	Dia
Kinematic	Linear	GateNet	E1	95	TC1	-0.151	1.226
Kinematic	Quad	GateNet	E1	95	TC1	-0.167	0.816
Kinematic	Linear	UNet+PnP	E1	95	TC1	-0.114	2.260
Kinematic	Linear	GateNet	E2	99	TC1	-0.061	2.452
Kinematic	Linear	GateNet	E1	99	TC1	-0.087	2.011
Kinematic	Linear	GateNet	E1	95	TC2	0.273	N/A
Aircraft	Linear	GateNet	E3	95	TA1	-0.306	2.136
Aircraft	Quad	GateNet	E3	95	TA1	-0.527	1.760
Aircraft	Linear	UNet+PnP	E3	95	TA1	-0.350	3.513
Aircraft	Linear	GateNet	E4	95	TA1	-0.454	4.289
Aircraft	Linear	GateNet	E3	99	TA1	-0.468	3.843
Aircraft	Linear	GateNet	E3	95	TA2	-0.495	3.857

Table 1. Incremental stability verification across different system configurations. The dynamics model (4D-Kinematic or 6D-Aircraft); the perception-contract form (linear or quadratic polynomial); the pose estimator (GateNet or UNet+PnP); environmental region (E1 and E3 are about 8x smaller than E2 and E4); the confidence level of the perception contract ($pr = 95, 99\%$); and the leader’s trajectory type (TC1 and TA1: mild maneuvers; TC2 and TA2: more aggressive maneuvers). See Section 5.1.3 for a detailed discussion of these results.

Table 1 summarizes these results and we can make a few observations: *Quadratic polynomial form perception contract generally produce tighter ultimately invariant sets than linear contracts*, as illustrated by lines 1–2 and 7–8. This is because the quadratic form reduces uncertainty in the relative pose estimates around critical target regions, resulting in less conservative abstractions for which stronger incremental stability certificates can be computed.

GateNet generally has tighter perception contracts than UNet+PnP, over the considered range of state and environment variations, (see lines 1–3 and 7–9 in Table 1). For the same contract form and confidence level, GateNet yields smaller uncertainty regions, reflecting more accurate relative pose estimates. However, this observation is not universal. Further, variations in network depth for GateNet and UNet may alter these results.

Larger environmental variations lead to larger ultimately invariant sets, as shown by lines 1–4 and 7–10. Expanding the environmental domain E introduces greater variability in images, requires larger perception contracts. The observed change in diameter Dia across different environmental regions (for example, $1.226 \rightarrow 2.452$, $2.136 \rightarrow 4.289$), represent some of the largest observed differences among all configurations, and they underscore the sensitivity of incremental stability to environmental variations.

Perception contracts with lower confidence levels (e.g., 95% vs. 99%) produce tighter ultimately invariant sets. A lower confidence level corresponds to a smaller bound on the set-valued estimation error, reducing diameter of ultimately invariant set for the abstract system. However, this comes at the cost of weaker probabilistic guarantees.

Finally, the complexity of the leader’s trajectory influences the size of the ultimately invariant set. More aggressive trajectories produce bigger sets, as they induce bigger deviations in the follower’s state, increasing over-approximation errors in the computation of γ and $C_{P,\mathcal{R}}$ and resulting in bigger contraction rates λ (less negative) and diameters Dia . For

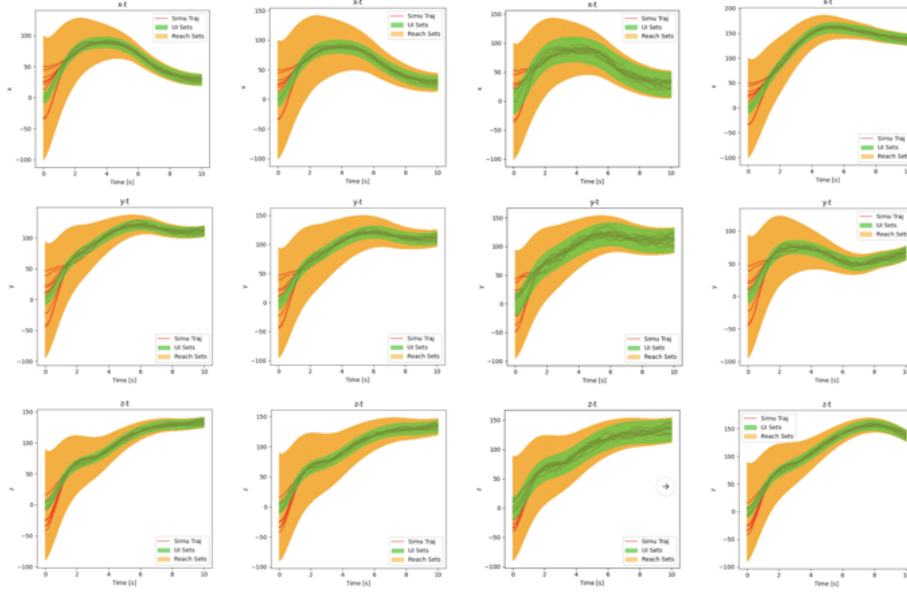


Fig. 6. Visualization of *Over-Approximated Reachable Sets* (orange), *Ultimately Invariant Sets* (green), and *Simulated Trajectories* (red) for translational coordinates x (top row), y (middle row), and z (bottom row) over time t . Columns (left to right): (1) GateNet-small-Mild, (2) UNet+PnP-small-Mild, (3) GateNet-big-Mild, (4) GateNet-small-Aggressive Maneuvers.

the 4D kinematic system, sufficiently aggressive leader trajectories can violate the assumptions required for incremental stability analysis, preventing any formal certification of stability in such case (Line 6 of Table 1).

5.1.4 Validating Contraction Analysis in Simulation. In this section, we validate our incremental stability analysis using Unreal Engine simulations under variations in the estimator, environmental region, and leader trajectory. Using the contraction rate λ and bound $C_{P,R}$ obtained from the abstract systems in Section 5.1.3 (setting contract form as linear, and confidence level as 95%), we generate over-approximated reachable sets—shrinking at rate λ —and the corresponding ultimately invariant sets. We then simulate trajectories of the concrete system (3) using sampled initial states, environmental factors, under two perception pipelines and various leader trajectories.

Fig. 6 shows that the computed ultimately invariant sets (green) tightly enclose the simulated trajectories (red) near the end of the horizon in most cases, and the trajectories remain within the over-approximated reachable sets (orange), supporting the soundness of our analysis. In some configurations, trajectories occupy only a subset of the invariant set, reflecting expected conservativeness; if trajectories were to deviate from the reachable sets, it would indicate that the perception contract does not sufficiently bound estimation uncertainty, suggesting the need for a higher confidence level.

5.2 Automated Landing with Lyapunov Perception Contracts

AutoLanding is a vision-based auto-landing system for fixed-wing aircraft. The system consists of a DNN-based pose estimator and a controller that uses the estimated pose to guide the aircraft along a reference trajectory with constant slope. The system dynamics is defined in terms of error dynamics with respect to this reference trajectory. AutoLanding-like systems are being developed for various types of aircraft to land in GPS-denied scenarios and airports without

expensive Instrument Landing Systems (ILS). The target set X^* is a bounded region where the error between aircraft state and reference state should be less than 3.6m [10]. For our analysis, we developed a detailed simulation of the aircraft dynamics, the controller, and the perception pipeline implemented with visual pose estimation.

5.2.1 AutoLanding Control System with DNN-based Pose Estimator. The state space X of the AutoLanding system is defined by the state variables $(\mathbf{x}, \boldsymbol{\theta})$, where $\mathbf{x} = (x, y, z)$, $\boldsymbol{\theta} = (\phi, \theta, \psi)$ (roll, pitch, yaw) are position and orientation of aircraft. The perception pipeline functions as an observer, denoted by h , which estimates the aircraft's positional states relative to the touchdown point. Specifically, the observation space Y is defined by the estimated position $(\hat{x}, \hat{y}, \hat{z})$. The orientation $\boldsymbol{\theta}$ are excluded from the observation space, as they are directly measured by onboard sensors, such as gyroscopes, rather than inferred from vision.

Dynamics. In this work, we use the simplified aircraft dynamics from [32]:

$$\begin{pmatrix} \dot{\mathbf{x}} \\ \dot{\boldsymbol{\theta}} \end{pmatrix} = \begin{pmatrix} b_1 & 0 \\ 0 & B_2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \quad (13)$$

where $u_1(t) = v(t)$ is the velocity of the aircraft and $u_2(t) = \mathbf{w}(t) = (w_x, w_y, w_z)$, represents the angular velocity of the aircraft and

$$b_1(\boldsymbol{\theta}) = (\cos \psi \cos \theta, \sin \psi \cos \theta, -\sin \theta)^T \quad (14)$$

$$B_2(\boldsymbol{\theta}) = \begin{pmatrix} 1 & \sin \phi \cos \theta & \cos \phi \sin \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi \sec \theta & \cos \phi \sec \theta \end{pmatrix} \quad (15)$$

A reference trajectory defined by reference position $\mathbf{x}_r$, reference orientation $\boldsymbol{\theta}_r$, reference velocity v_r and reference angular velocity $\mathbf{w}_r$ is given to guide aircraft to safely land toward the runway. This reference approach trajectory begins 2.3 kilometers from the touchdown point at an attitude of 120 meters, with a descend slope of -3° in the z direction, and maintains a constant speed of 10m/s.

Lyapunov Function and Controller. Denote $B_{2r} = B_2(\boldsymbol{\theta}_r)$ be B_2 matrix taking reference angles; R be the rotation matrix from the local frame to inertial frame. The error state is derived as $\mathbf{x}_e = R^T(\mathbf{x}_r - \mathbf{x})$ for positional error and $\boldsymbol{\theta}_e = \boldsymbol{\theta}_r - \boldsymbol{\theta}$ for rotational error.

We adopt the control policy from [32] by replacing actual error states (x_e, y_e, z_e) with the observed error states $(\hat{x}_e, \hat{y}_e, \hat{z}_e)$ as:

$$u_1 = v_r \cos \psi_e \cos \theta_e - \gamma^2 \hat{x}_e \quad (16)$$

$$u_2 = \mathbf{w}_r + B_{2r}^{-1} \{ (B_{2r} - B_2) \mathbf{w}_r + p_e + \hat{q} \} \quad (17)$$

where $\gamma = 2$, $p_e = (k_1 \sin \phi_e, k_2 \sin \theta_e, k_3 \sin \psi_e)^T$, $\mathbf{k} = (k_1, k_2, k_3)^T = (5, 5, 5)^T$ and $\hat{q} = (0, -\frac{\hat{z}_e v_r}{k_2}, \frac{\hat{y}_e v_r \cos \theta_e}{k_3})^T$.

We consider the following Lyapunov function:

$$V = \frac{1}{2} \mathbf{x}_e^T \mathbf{x}_e + \mathbf{k}^T f(\boldsymbol{\theta}_e) \quad (18)$$

where $f(\boldsymbol{\theta}_e) = (1 - \cos \phi_e, 1 - \cos \theta_e, 1 - \cos \psi_e)^T$. From the requirement that the error between aircraft state and reference state is less than 3.6m, we formally define $X^* = \{(\mathbf{x}_e, \boldsymbol{\theta}_e) | V(\mathbf{x}_e, \boldsymbol{\theta}_e) \leq 6.48\}$ and choose $W(\mathbf{x}_e, \boldsymbol{\theta}_e) = 0.001$ for all $(\mathbf{x}_e, \boldsymbol{\theta}_e) \in X \setminus X^*$.

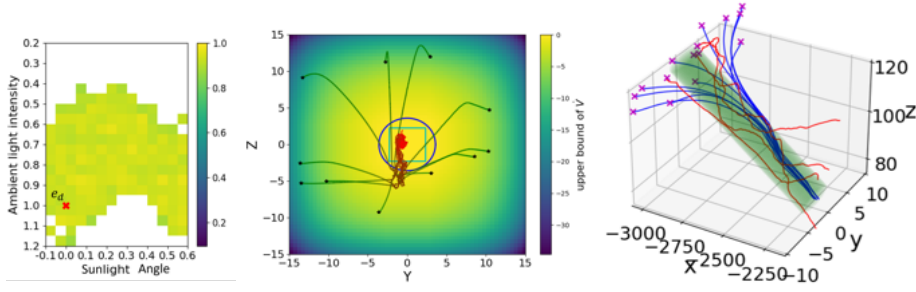


Fig. 7. (a): Colored region represent operational design domain identified by DaReLPC. (b): Heat map of upper bound of Lyapunov function's time derivative. Simulated trajectories under $\tilde{E}$ (green-red) converge to region with non-negative Lyapunov time derivative (cyan rectangle) contained in X^* (blue circle). (c): Simulation trajectories under $\tilde{E}$ (blue) converge to X^* (green tube), trajectories under $E \setminus \tilde{E}$ (red) is unstable.

Vision-based Observer. The observations $(\hat{x}, \hat{y}, \hat{z})$ are computed using a realistic perception pipeline composed of three functions: (1) The airplane's front camera generates an image. (2) A key-point detector detects specific key-points on the runway in that image using U-Net [38]. In this case study, we use the U-Net implementation from [1]. (3) Pose estimator uses the detected key-points and the prior knowledge of the actual positions of those key-points on the ground to estimate the pose of the aircraft. This last calculation is implemented using the perspective-n-point algorithm [13]. Apart from the impact of the environment on the image, the observer is deterministic.

Environment. We consider two different environments: 1) a simulated environment rendered using Gazebo and 2) Google Earth data for Heathrow and O'Hare airports pulling using CesiumJS. For the Gazebo simulation, we study the effects of the ambient light intensity in $[0.2, 1.2]$ and the position of the sun, represented by a spotlight with angle $[-0.1, 0.6]$. The effect of different environmental parameters is shown in Figure 7. These ranges are chosen to include a nominal environment $e_d = (1.0, 0)$. For the data from Google Earth, we choose ambient light intensity between $[0.6, 1]$ and snow level $[0, 0.4]$ as E_0 . The effect of these environmental parameters is shown in Figure 8. The nominal environment in this case is $e_d = (1.0, 0.0)$ corresponding to no change in lighting and no snow.

5.2.2 Safety Analysis of AutoLanding with Gazebo Data. We first apply DaReLPC on the AutoLanding system for data from Gazebo simulator. The algorithm outputs an environmental condition $\tilde{E} \subset E$ and a corresponding $(\tilde{E}, X^*)$ -Lyapunov perception contract $M_{\tilde{E}}$, after performing 12 rounds of refinements. The resulting $\tilde{E}$ is visualized in Figure 7. In the figure, the white region represents the environmental parameters that are removed (not provably stable) during shrinking. For the colored region, different colors represent the empirical conformance of h to M over each environmental parameter partition.

Computed perception contract achieves target conformance measure. To compute the perception contract, we set target conformance measure $pr = 90\%$, the gap between actual conformance and empirical conformance $\epsilon = 1\%$, and the confidence parameter $\delta = 0.1\%$. The empirical conformance measure should be at least 91% for approximately 34K samples, which we round up to 40K. The resulting perception contract $M_{\tilde{E}}$ achieves a 92.4% empirical conformance measure over $\tilde{E}$. Furthermore, 30 random simulations under $\tilde{E}$ yield a 98.6% empirical conformance measure, demonstrating that $M_{\tilde{E}}$ not only meets but can exceed the required conformance for certain realistic sampling distributions.

DaReLPC finds nontrivial ODD. As shown in Figure 7 (a), when the ambient light intensity is below 0.6 or above 1.15, the perception performance degrades, and these parts of the environment E removed partitions under these



Fig. 8. Heathrow and O'Hare International runway images pulled using CesiumJS under different environmental conditions.

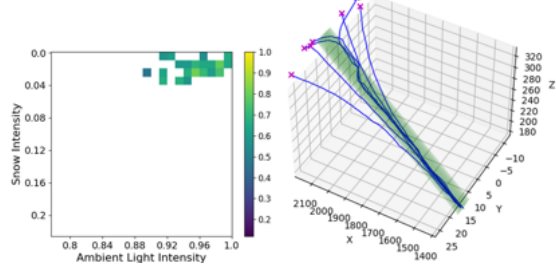


Fig. 9. (a): Identified ODD for Google Earth data. (b): Simulated trajectories under $\tilde{E}$ converge to X^* (green tube).

conditions. Interestingly, around $[0.95, 1.1] \times [0.25, 0.4]$ the ambient conditions are near nominal, yet the perception performance is poor. This can be explained as the glare from the low sun overwhelms the keypoint detector. Around $[0.4, 0.6] \times [0.25, 0.4]$, on the other hand, the sun's light compensates for low ambient light, maintaining reliable perception performance for landing.

Correctness and system-level stability. As shown in Fig 7 (b), the blue circle with a radius of 3.6 represents the target set X^* on $y - z$ subspace. The color bar indicates that all states in $X \setminus X^*$ has a negative value of Lyapunov function's time derivative, meaning that any trajectory starts from $X \setminus X^*$ will converge into X^* . We run 10 simulations under $\tilde{E}$, all simulated trajectories (shown as green-red lines) converge to the target set X^* (the blue circle). In Figure 7 (c), the same trajectories (blue lines) converge into X^* (the green tube); while the sampled trajectories under invalid environment $E \setminus \tilde{E}$ (red lines) are not guaranteed to converge into X^* .

5.2.3 Safety Analysis of AutoLanding on Google Earth Data. To further show generality of our analysis, we analyze AutoLanding with data generated from Google Earth for actual airports. We use the same X^* used for Gazebo data. The result is shown in Figure 9. The identified range of environment that AutoLanding is stable is shown in Figure 9 (a). The perception pipeline is sensitive on Google Earth data and small perturbation from the environment can drastically change the perception result. The identified environment is intuitive: as the ambient light reduces or the level of snow increases, we can no longer proof stability of the system for X^* and the system will converge to a much larger set or even becomes unstable. In this paper, we analyze the perception pipeline as is, leaving its improvement based on our analysis for future work. Figure 9 (b) shows random simulated trajectories under $\tilde{E}$ (Blue trajectories). We can see under $\tilde{E}$, all simulations properly converge to X^* . However, under $E \setminus \tilde{E}$, the system can deviate from X^* .

5.2.4 Convergence and Conformance of Perception Contracts. For termination of Algorithm 3, we assume that the perception contract we computed is convergent, i.e. as candidate contracts are computed for a sequence of environments converging to e_d , the contracts also converge to h (pointwise). To test this, we pre-sampled 80k data points and set $pr = 85\%$, $\delta = 0.1\%$. We reduce $\tilde{E}$ for $M_{\tilde{E}}$ by invoking ShrinkEnv function 0, 3, and 6 times on the set of environments with pre-sampled data points. ShrinkEnv reduces the range of environments by removing environmental parameters where the observations is far from the actual states and removing corresponding data points sampled under these environments. The result is shown in Figure 10. We observe that after 0,3,6 round of shrinking $h(x, e_d) \in M_{\tilde{E}_6}(x) \subseteq M_{\tilde{E}_3}(x) \subseteq M_{\tilde{E}_0}(x)$, which provides evidence that the perception contract is convergent.

As we reduce the range of environmental parameters with 0, 3 and 6 shrinking, the training data set reduced from 80000 to 71215 and 62638, respectively. Therefore, according to Proposition 5, the empirical conformance measure

should at least be 85.8% for all these three cases to achieve the 85% actual conformance measure. In practice, we are able to get 85.9%, 87.5% and 89.1% empirical conformance measure. From this, we can see that refinement preserves the conformance measure of the perception contract.

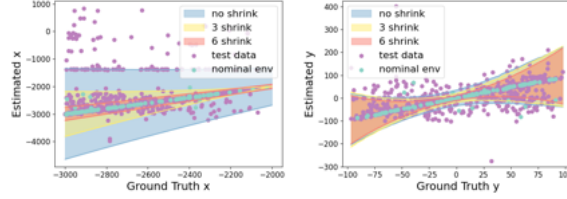


Fig. 10. Visualizing perception contract with different number of shrinking for different dimensions. Perception contract convergence for the x-dimension (left) and y-dimension (right), showing estimated state versus ground truth. In each plot, the blue, yellow, and red shaded regions represent the perception contract bounds after 0, 3, and 6 rounds of environment shrinking respectively. With increase in shrinking rounds, the contract bounds tighten, confirming that the perception contracts converge as the environment narrows toward the nominal condition.

6 Conclusion

In this paper, we presented *incrementally stable perception contracts (ISPC)*, a unified framework for verifying incremental stability of vision-based autonomous systems through perception contracts and contraction analysis. Perception contracts abstract the perception pipeline as a set-valued map over a specified environment, enabling modular reasoning about closed-loop reliability without requiring differentiable models of the learned observer. As a special case, when the system admits an equilibrium, the framework specializes to asymptotic stability via Lyapunov perception contracts (LPCs), which enable operating design domain (ODD) identification in finitely parameterized environments.

We developed efficient methods for constructing perception contracts from data, computing certified contraction rates via bisection over semidefinite feasibility problems, and bounding ultimately invariant tube diameters. These tools provide rigorous guarantees on trajectory convergence under visual uncertainty and, in the LPC setting, identify the largest environmental region over which asymptotic stability can be formally certified.

We validated the framework on two realistic vision-based control tasks. For formation flying with 6D rigid-body aircraft dynamics, quadratic perception contracts tighten certified tubes by 30–35% over linear contracts, and GateNet-based estimators achieve faster verified convergence than UNet+PnP under identical dynamics and environments. For automated landing, LPCs enable practical ODD identification, certifying safe landing conditions under varying lighting and weather while excluding configurations that violate stability guarantees. Across both tasks, system configuration directly shapes certificate tightness: aggressive trajectories and larger domains increase tube diameter and drive contraction rates toward instability, demonstrating perception contracts’ value in balancing operational flexibility with formal safety guarantees.

Future work includes generating state-dependent contraction metrics $P(x)$ to improve local tightness, exploring richer contract forms (e.g., neural network over-approximations), and extending the framework to identify unsafe environmental regions through systematic falsification. Integration with runtime monitoring and adaptive control would enable autonomous systems to dynamically adjust behavior as environmental conditions approach ODD boundaries, providing both offline certification and online assurance for vision-based autonomy in safety-critical applications.

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